INVERTIS UNIVERSITY BAREILLY

# CENTER FOR SKILL AND ENTREPRENEURSHIP DEVELOPMENT (CSED)

## INDUSTRIAL INTERNET OF THINGS

PROGRAM CODE: **IIOT-4**

PROGRAM NAME: **MACHINE LEARNING**

PROJECT NAME: **DEMAND FORECASING AND INVENTORY OPTIMIZATION**

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COURSE NAME: **B.TECH(ARTIFICAL INTELLIGENCE)**


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# Introduction:

Demand forecasting and inventory optimization are critical aspects of supply chain and operations management. In any business that deals with products—whether retail, manufacturing, or e-commerce—it is essential to maintain a balance between supply and demand. If a company maintains too much inventory, it incurs high holding costs such as storage, insurance, and risk of damage or obsolescence. On the other hand, if inventory is too low, it leads to stockouts, lost sales, and customer dissatisfaction.

Traditionally, businesses relied on manual forecasting methods such as historical averages or human intuition. However, these methods are often inaccurate because they cannot capture complex patterns such as seasonal variations, trends, or sudden changes in demand. With the growth of data and computational power, **machine learning techniques** have become an effective solution to improve forecasting accuracy.

This project focuses on using two fundamental machine learning algorithms:

- **Linear Regression** for predicting future demand values
- **Logistic Regression** for making inventory-related decisions such as whether to reorder stock

By combining prediction and decision-making, this system provides a complete solution for inventory management. The integration of statistical models with business logic enables organizations to reduce costs, improve efficiency, and enhance customer satisfaction.

# Problem Statement:

Many organizations face significant challenges in managing their inventory efficiently. The major problems include inaccurate demand forecasting, poor inventory planning, and lack of automation in decision-making processes. These issues lead to both financial losses and operational inefficiencies.

One of the key problems is **overstocking**, where businesses store more products than required. This results in increased holding costs, wastage (especially for perishable goods), and blocked capital. On the other hand, **understocking** leads to stockouts, which directly impacts customer satisfaction and brand reputation.

Another major issue is that traditional forecasting methods fail to consider dynamic factors such as seasonal demand, trends, and external influences. Moreover, many small and medium businesses still rely on manual decision-making, which is time-consuming and prone to errors.

Therefore, there is a need for an intelligent system that can:

- Predict demand accurately using historical data
- Automate inventory decisions
- Optimize stock levels to reduce costs

This project aims to address these challenges using machine learning-based techniques.

# Objective of the Project:

The primary objective of this project is to design and develop a smart system for demand forecasting and inventory optimization using machine learning algorithms.

The specific objectives are:

- To analyze historical sales data and identify patterns in demand
- To develop a predictive model using **Linear Regression** for estimating future demand
- To build a classification model using **Logistic Regression** to decide whether to reorder stock
- To implement inventory optimization techniques such as EOQ, Safety Stock, and Reorder Point
- To reduce overall inventory-related costs including holding and ordering costs
- To improve decision-making efficiency by automating the process
- To provide insights that help businesses plan inventory more effectively

This project aims to bridge the gap between theoretical machine learning concepts and practical business applications.

# Methodology:

The methodology of the project follows a systematic and structured approach, consisting of multiple stages:

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## Data Collection

The first step involves collecting historical sales data. This data typically includes information such as date of sale, quantity sold, product details, and inventory levels. The accuracy of the model depends heavily on the quality of this data.

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## Data Preprocessing

Raw data is often incomplete or inconsistent. Therefore, preprocessing is required to clean and prepare the data. This includes handling missing values, removing duplicate entries, and converting data into a suitable format. For example, date values are converted into numerical features such as day, month, and year.

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## Feature Engineering

Feature engineering is an important step where new features are created from existing data to improve model performance. Examples include extracting seasonal patterns, identifying trends, and creating lag variables based on previous sales.

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## Model Building: Linear Regression

Linear Regression is used to predict future demand based on historical data. It establishes a relationship between independent variables (such as time) and dependent variables (sales).

The model is trained using past data and then used to predict future values.

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## Model Building: Logistic Regression

Logistic Regression is used for classification. It helps in deciding whether the stock should be reordered or not based on predicted demand and current inventory levels.

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Inventory Optimization

After prediction and classification, inventory optimization techniques are applied. These include:

Economic Order Quantity (EOQ)

Safety Stock

Reorder Point

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Visualization and Output

Finally, the results are presented in a clear and understandable format using graphs, tables, or dashboards. This helps users make informed decisions.

# Literature Review:

Various studies have shown that machine learning techniques significantly improve demand forecasting accuracy. Traditional time-series models like moving averages and ARIMA have limitations in handling complex patterns.

Recent research highlights:

- ML models can capture hidden patterns in data
- Regression models are simple yet effective for forecasting
- Classification models help automate decision-making

This project combines both regression and classification to provide a complete solution.

Together, these studies provide a strong foundation for the design and implementation of a **Smart Indoor Energy Optimization System**, combining sensor-based automation with real-time cloud monitoring.

# Explanation of Project (Working):

The working of the project can be explained step-by-step as follows:

Initially, the system takes historical sales data as input. This data is then preprocessed to remove inconsistencies and convert it into a usable format. After preprocessing, feature engineering is performed to extract meaningful information from the data.

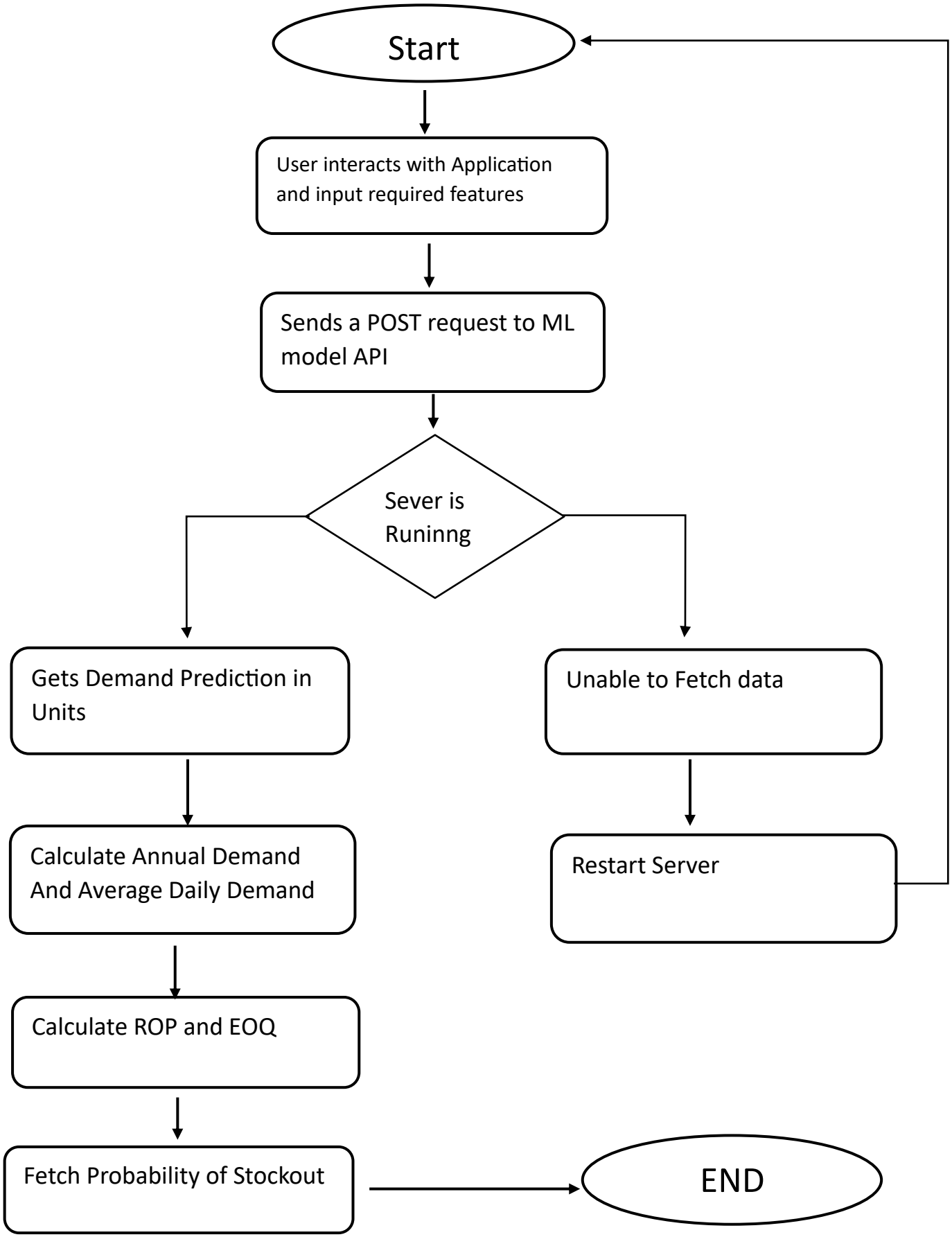
Next, the Linear Regression model is trained using the processed data. This model learns the relationship between time and sales, allowing it to predict future demand values. These predictions help in understanding how much stock will be required in the future.

After predicting demand, the Logistic Regression model is used to make decisions regarding inventory. It takes inputs such as predicted demand and current stock levels and outputs whether the stock should be reordered or not.

Once the decision is made, inventory optimization techniques are applied. EOQ determines the optimal quantity to order, Safety Stock ensures buffer inventory, and Reorder Point identifies the right time to reorder.

Finally, the system displays the results, including predicted demand, reorder decisions, and optimized inventory levels.

# Work Flow :


Start
User interacts with Application and input required features
Sends a POST request to ML model API
Sever is Runinng
Gets Demand Prediction in Units
Unable to Fetch data
Calculate Annual Demand And Average Daily Demand
Restart Server
Calculate ROP and EOQ
Fetch Probability of Stockout
END

# Program :

**1.Code to train demand model(Linear Regression):**

```
import pandas as pd
import joblib
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Load dataset
df = pd.read_csv("Superstore.csv")

# Create Month column
df["Month"] = pd.to_datetime("2024-01-01") + pd.to_timedelta(range(len(df)),
unit="D")
df["Month"] = pd.to_datetime(df["Month"]).dt.month

# Create Unit Price
df["Unit_Price"] = df["Sales"] / df["Quantity"]

# Aggregate monthly demand by product
monthly = df.groupby(["Month","Sub-Category"]).agg({
    "Quantity":"sum",
    "Unit_Price":"mean"
}).reset_index()
```

```
# One-hot encode Sub-Category
monthly = pd.get_dummies(monthly, columns=["Sub-Category"])

# Define X and Y
X = monthly.drop("Quantity", axis=1)
y = monthly["Quantity"]

# Train/Test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

# Prediction
pred = model.predict(X_test)

# Evaluate
print("R2 Score:", r2_score(y_test, pred))

# Save model
joblib.dump(model, "demand_model.pkl")

print("Model saved successfully")
```

**Code to train Stockout model(Logistic Regression):**

```
import pandas as pd
import joblib
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

df = pd.read_csv("data/SampleSuperstore.csv")

# create artificial date
df["Order_Date"] = pd.date_range(
    start="2022-01-01",
    periods=len(df),
    freq="D"
)

df["Month"] = df["Order_Date"].dt.month

# unit price
df["Unit_Price"] = df["Sales"] / df["Quantity"]

# simulate inventory level
df["Current_Stock"] = np.random.randint(50,500,len(df))
```

```
# stockout logic
df["Stockout"] = df["Quantity"] > df["Current_Stock"]

# FEATURES (match dashboard)
X = df[["Quantity","Current_Stock"]]

y = df["Stockout"]

X_train,X_test,y_train,y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

model = LogisticRegression()

model.fit(X_train,y_train)

pred = model.predict(X_test)

print("Accuracy:",accuracy_score(y_test,pred))

joblib.dump(model,"models/stockout_model.pkl")
```

```
print("Stockout model saved")
```

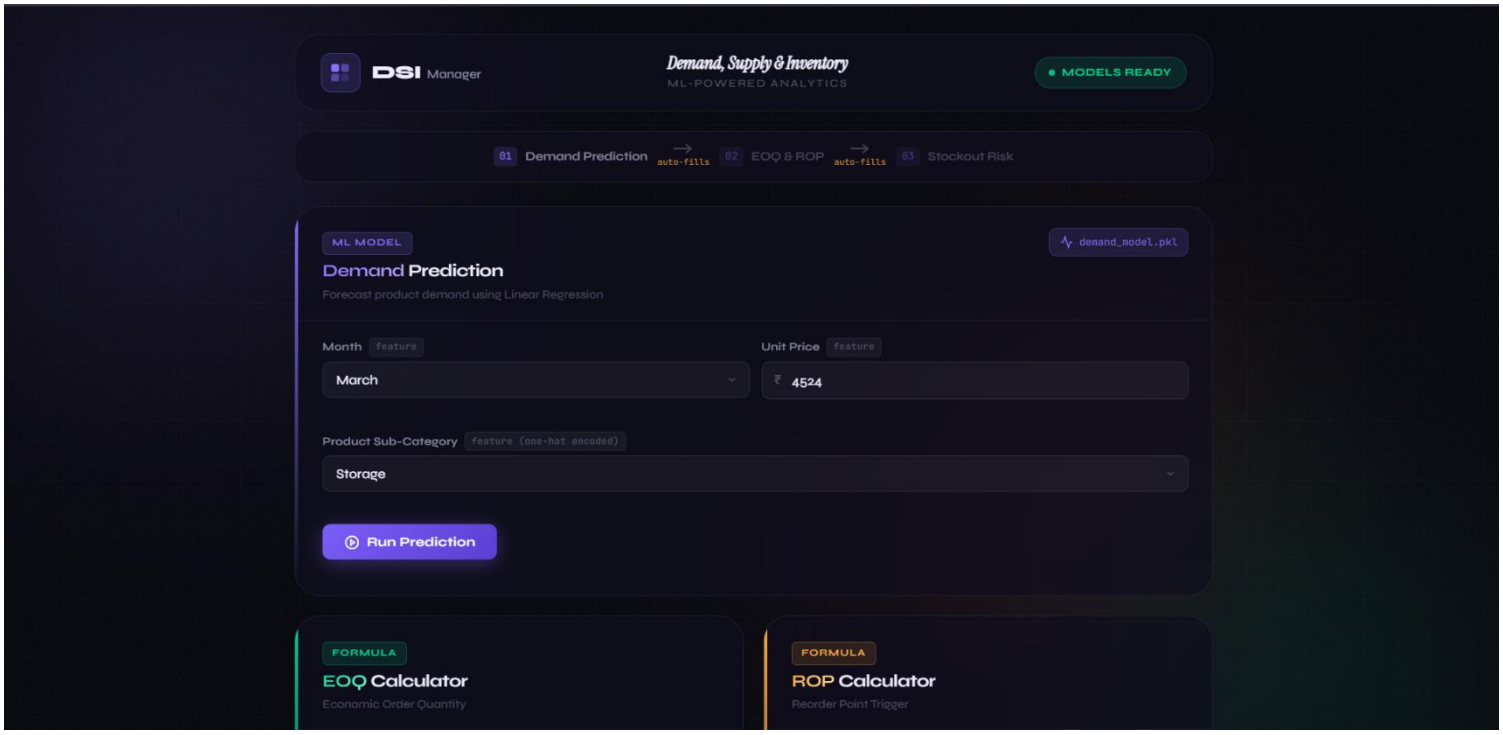
**Code to Create API of ML models:**

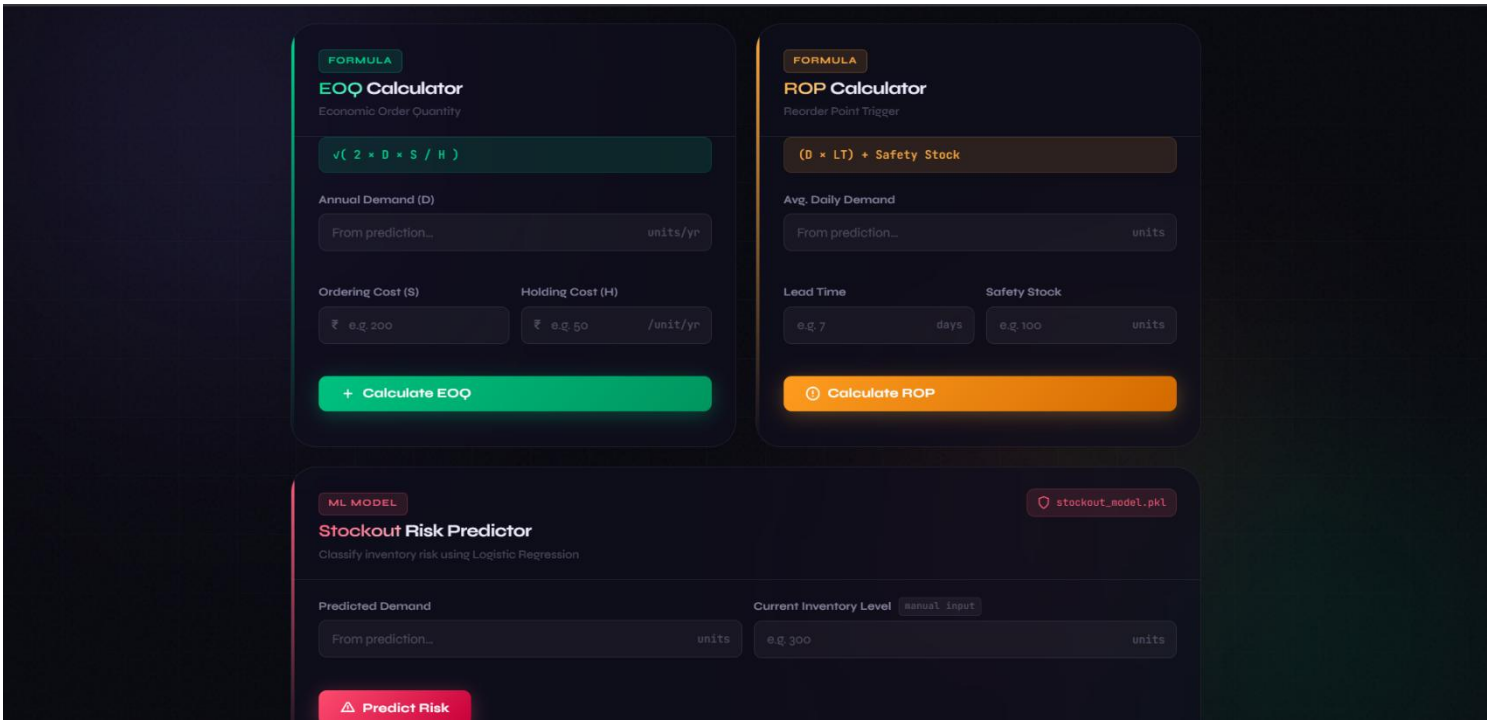
```
from flask import Flask, request, jsonify
from flask_cors import CORS
import joblib
import numpy as np

app = Flask(__name__)
CORS(app)  # Allow requests from the frontend (restrict origins in production)

# — Load models at startup —
linear_model  = None
logistic_model = None

try:
    linear_model  = joblib.load('models/demand_model.pkl')
    logistic_model = joblib.load('models/stockout_model.pkl')
    print("Models loaded successfully.")
except FileNotFoundError as e:
    print(f"Warning: Could not load model — {e}")
    print("Ensure demand_model.pkl and stockout_model.pkl are in the models/
directory.")

def validate_request(data, expected_len=None):
    """Returns (features_array, error_string).  error_string is None on success."""
```

```
    if not data or 'features' not in data:
        return None, "Request body must be JSON with a 'features' key."
    features = data['features']
    if not isinstance(features, list):
        return None, "'features' must be a list."
    if expected_len and len(features) != expected_len:
        return None, f"Expected {expected_len} features, got {len(features)}."
    return np.array(features, dtype=float).reshape(1, -1), None

# — POST /predict_demand
————————————————————————————————————————————————————————————

@app.route('/predict_demand', methods=['POST'])
def predict_demand():
    if linear_model is None:
        return jsonify({'error': 'Demand model not loaded. Check
models/demand_model.pkl'}), 503

    features, err = validate_request(request.get_json(), expected_len=19)
    if err:
        return jsonify({'error': err}), 400

    prediction = linear_model.predict(features)
    return jsonify({
        'model_type': 'Linear Regression',
        'prediction': float(prediction[0])
    })
```

```
# — POST /predict_risk
_____________________________________________________________________

@app.route('/predict_risk', methods=['POST'])
def predict_risk():
    if logistic_model is None:
        return jsonify({'error': 'Stockout model not loaded. Check
models/stockout_model.pkl'}), 503

    features, err = validate_request(request.get_json(), expected_len=2)
    if err:
        return jsonify({'error': err}), 400

    prediction  = logistic_model.predict(features)
    probability = logistic_model.predict_proba(features).tolist()
    return jsonify({
        'model_type': 'Logistic Regression',
        'prediction': int(prediction[0]),   # 0 = Low Risk, 1 = High Risk
        'probability': probability          # [[p_low, p_high]]
    })

if __name__ == '__main__':
```

# Snippets of UI:



# Result & Discussions :

The results of the project show that the machine learning models are effective in predicting demand and optimizing inventory.

The Linear Regression model successfully captures the trend in sales data and provides reasonably accurate predictions. The Logistic Regression model effectively classifies whether stock needs to be reordered, helping automate decision-making.

The use of inventory optimization techniques further improves efficiency by minimizing costs and ensuring availability of products.

However, the performance of the models depends on the quality and quantity of data. In cases where data is limited or highly non-linear, the accuracy may decrease. Advanced models can be used to improve performance in such scenarios.

# Future Scope:

The project can be further enhanced in several ways. Advanced machine learning models such as Random Forest, Gradient Boosting, or LSTM can be used to improve prediction accuracy. Real-time data integration can be implemented to make the system more dynamic and responsive.

The system can also be deployed on cloud platforms to make it scalable and accessible from anywhere. Integration with ERP systems can help automate the entire supply chain process.

Additionally, the use of IoT devices can enable real-time inventory tracking, further improving efficiency.

# Conclusion:

In conclusion, this project demonstrates the application of machine learning in solving real-world business problems related to demand forecasting and inventory management.

By using Linear Regression, the system is able to predict future demand, while Logistic Regression helps in making intelligent inventory decisions. The integration of these models with inventory optimization techniques ensures efficient stock management.

Overall, the project provides a cost-effective and efficient solution for businesses, helping them improve performance and customer satisfaction.